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A Comparative Organizational Study of Performance and Size Correlates in Inpatient Psychiatric Departments

Lawrence G. Hrebiniak
and Joseph A. Alutto

In this study of 338 inpatient psychiatric departments five hypotheses concerning organizational performance and size are tested and supported. It is found that psychiatric inpatient department performance varies with hospital ownership and is inversely related to department size. It is also found that increases in department size are not marked by changes in the relative size of administrative components. With increases in size, however, private hospitals tend to employ nonprofessional or paraprofessional support staff, while public hospitals are marked by a proliferation of professional staff. Furthermore, as inpatient departments increase in size they tend to increase the number of boundary spanning and buffering structures in existence. Discussed is the extent to which size and organizational ownership affect the performance of the psychiatric department.

In recent research on the characteristics of institutions treating mental illness, reference has repeatedly been made to the displacement of therapeutic goals, the lack of communication and integration of specialized activities, the increasing level of bureaucratization, and the consequent deleterious effect of these and other factors on the efficiency and effectiveness of the treatment organization (Goffman, 1961; Caudill, 1958; Greenblatt et al., 1957; Stanton and Schwartz, 1954; Cumming and Cumming, 1956 and 1962; Salisbury, 1962).

Examination of the empirical and theoretical bases of these studies indicates that some important organizational and system variables have been neglected. The bulk of research, for example, has focused primarily on the mental hospital, ignoring such intermediate points in the health care system as the psychiatric departments of general hospitals. Also most organizations studied in previous research were large state facilities, indicating that the effects of potentially important variables such as size and ownership were not considered because they were, in effect, held constant. Therefore, previous findings could have been in large part representative of only a certain type of organization. In research on general hospital settings investigating the relationship between technology and structure or focusing on the operational rationality of medical care systems (Coser, 1958; Rosengren and DeVault, 1963), little work has been done on the technology and rationality of mental health institutions and psychiatric departments within general hospitals. Indeed, as Perrow (1965) stressed, much research has called for alterations in structure that are not viable and probably result from a lack of understanding of treatment technology and its relationship to organizational effectiveness.

The present study investigates the performance of the psychiatric inpatient departments of 338 general hospitals. Hospital ownership and department size are considered for their effects on department characteristics as well as performance. Some correlates of department size are discussed, including the relationship between size and (1) staffing patterns, (2) the relative size of administrative and professional staff, and (3) the development of input and output buffers for the treatment technology.

BACKGROUND AND HYPOTHESES

Like the mental hospital, the psychiatric department of a general hospital is characterized by an intensive treatment technology, which requires the hospitalization and isolation of

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the client (Thompson, 1967; Goffman, 1957). However, the large hospital devoted solely to the care of mental illness represents a final point in the total treatment system (Perrow, 1965), whereas the psychiatric department represents an intermediate point. Having developed as a subsystem of the short-term, acute-disease general hospital, the inpatient psychiatric department can be viewed first as providing treatment for short-term, acutely ill patients and, second, as channeling the remaining cases to other treatment locations. The goal of the department is not only treatment but also diagnosis and the attendant boundary-spanning function for the movement and care of patients (Baker, 1970; Baker and Schulberg, 1970).

Ownership and Performance

Thompson (1967) and others have indicated that environmental constraints on organizations differ as a function of ownership or sponsorship. In the mental health treatment system, government-owned hospitals typically have little choice of clientele, whereas private institutions can be very selective in their admissions policy (Belknap, 1956). This suggests that the boundary-spanning function of psychiatric departments in public general hospitals would be restricted if there were a lack of beds in their most likely "receiving" institutions, government-owned mental hospitals. Also, Stanley (1963) found that the socio-economic status of clients in private hospitals is generally perceived to be more favorable than that of clients in public municipal general hospitals, pointing to a possible relationship between the performance of the psychiatric department and hospital ownership.

Some research has indicated that the ownership of general hospitals has a significant impact on many aspects of their performance, including their efficiency and effectiveness, because of such intervening factors as type of constraints on resources, administrative style and control procedures, level of bureaucratization, and related institutional factors (Elling and Halebsky, 1961; Blau and Scott, 1962; Stanley, 1963; Fottler, 1971). For example, the autonomy of administrators and general staff is somewhat more limited in government-controlled organizations than in private organizations (Peabody and Rourke, 1965). Operating a hospital and making policy to serve private interests, especially profit-making ones, and performing a service solely in the public interest perhaps have different implications for department efficiency. Moreover, the extensive control mechanisms and high levels of bureaucratization generally found in public organizations can influence the performance of psychiatric departments. The hypothesis follows:

Hypothesis 1. The performance of the inpatient psychiatric department varies with hospital ownership, with departments in private hospitals performing better than departments in government-owned hospitals.

And in the light of increasing government services, the following hypothesis can be made:

Hypothesis 2. Inpatient psychiatric departments in private hospitals are smaller than those in government-owned hospitals.

Correlates of Department Size

Organizational size has been studied for its relationship with individual attitudes and behavior (Indik, 1965; Talacchi, 1960; Wilken, 1971; Ingham, 1970), technology and

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intraorganizational differentiation (Blau, 1970; Woodward, 1958), and the relative growth of administrative and technical staff (Anderson and Warkov, 1961; Terrien and Mills, 1955; Rushing, 1967; Klatzky, 1970; Bendix, 1956). Little work, however, has been done on the correlates of size of departments or subunits within larger organizational settings, as in the present study. Consideration of the correlates of department size should also provide needed (Perrow, 1965) insights into change and growth processes in health care organizations.

Administrative and professional staff. Anderson and Warkov (1961), in their study of Veterans Administration hospitals, found that the relative size of the administrative staff decreased as organizational size increased, contradicting the Parkinsonian proposal that increases in size would be marked by a proliferation of administrative staff relative to other personnel (Terrien and Mills, 1955). Within the intensive technology of the psychiatric department (Thompson, 1967; Hunt, 1970), these data suggest that departmental growth is marked by an increasing professionalization of staff. Whereas increased size might lead to a reduction in the relative size of the administrative staff, departmental growth may be associated with demands on the technology of professional-client interaction, which would increase the need for professional or technical employees (Bendix, 1956).

Another possibility not treated by Anderson and Warkov (1961) or other authors is that many of the administrative tasks in organizations with professionals can be assumed by professional employees rather than administrative staff. Decreases in the relative size of administrative staff, then, might not necessarily indicate economies of scale, but may merely reflect the placement of professional employees in positions where their treatment competencies are wasted. The increased professionalization of psychiatric departments with no increase, or a decrease, in department performance could denote such an inefficiency and the misuse of professional knowledge and other scarce resources. The following hypotheses can be made:

Hypothesis 3. The larger the inpatient psychiatric department, the smaller the relative size of the administrative staff and the larger the relative size of the professional staff.

Hypothesis 4. The performance of the inpatient psychiatric department is inversely related to department size.

While it is expected, then, that department size varies as a function of hospital ownership, it is also assumed that department performance is inversely related to size regardless of organizational ownership or sponsorship.

Buffers and structural change. Most organizations and functional departments facing uncertain task environments try to smooth environmental influences by surrounding their core technologies with input and output buffers (Thompson, 1967). On the input side, for example, psychiatric departments in a heterogeneous environment might try to identify and treat only homogeneous segments of their environment, thereby increasing the specialization or number of structural subunits within the department. Or structural units might be developed on the output side to facilitate the flow of patients out of the inpatient department. Given the interdependencies among organizations within the treatment system, the development of boundary-spanning buffers appears crucial to the performance of the psychiatric department. It seems probable that buffering

activities are especially important as the size of the psychiatric department increases. If size is positively related to the proliferation of professional staff and an increase in complexity or problems of coordination, then activities are needed to smooth input and output transactions, and the following hypothesis can be made:

Hypothesis 5. The existence of input and output buffers is positively related to size of the inpatient psychiatric department.

Data and Sample

From approximately 750 general hospitals in the 1969 Inventory of Psychiatric Services of the National Institute of Mental Health which met the criteria of the Institute for the provision of inpatient psychiatric services physically and technologically separate from regular medical and surgical care, 338 inpatient departments were used in the present analysis. Data from the remaining departments were not fully complete in the Inventory; because of the limitations of secondary data collection, the absence of any variable was considered sufficient reason to eliminate the department from consideration. The departments represented governmentally controlled city and county ($n = 47$) and Veterans Administration hospitals ($n = 43$), and both profit ($n = 76$) and nonprofit ($n = 172$) private general hospitals. Examination of the departments of the sample revealed no significant divergence from the total available population with respect to ownership, department size, and geographical location.

Precautions were taken by the Survey and Reports Section of the National Institute of Mental Health to ensure the reliability and accuracy of data in the Inventory. Checks were made to determine inter-item consistency or item validity, for example, by examining the relationship between number of paid staff and staff salaries or between the present data and previous records. Nonetheless, it was recognized that data derived from the Inventory were in most cases compiled by administrators who may not have been as concerned as most social scientists about accuracy and consistency. However, since original collection of data from such a large number of psychiatric departments would entail a great deal of time and expense, and since the present research represents an exploratory investigation, it was decided to use the Inventory data.

Variables

Department size. Size of department was measured by the number of beds devoted solely to inpatient psychiatric care. This measure, taken directly from the Inventory of Psychiatric Services, represents a common classification of size in previous research and in descriptions of general hospital characteristics (Linn, 1970; Perrow, 1965; Ullman, 1967). To the extent that the Veterans Administration hospitals studied by Anderson and Warkov (1961) were operating at full or nearly full capacity, the present measure of size would be identical to their measure of average daily patient census. The present high demand for hospital medical, surgical, and psychiatric services also suggests that patient census is, in fact, limited only by bed capacity, that is, organizational size in the present study.

Department performance. Three measures of performance were used. First, because of the treatment and distribution goals of the inpatient department, a weighted discharge rate was developed. The total number of discharges from each inpatient department for the year 1969 was divided by

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department size to provide a utilization or turnover ratio of output per bed. Data on re-admission rates were not available, thus making it impossible to adjust the gross discharge figure. This measure adjusts for size differences, while a time period of sufficient duration was used to neutralize short-term fluctuations in the number of discharges.

It is recognized that discharge rate is not necessarily a measure of department effectiveness or quality of care. It is, however, a useful variable for the present research. First, discharge rate suggests the relative efficiency of the psychiatric department in utilizing resources and performing its boundary-spanning function. If for each type of hospital, for example, it can be shown that size of the department affects discharge rate negatively, this could signal variations in efficiency as a function of size. Because of the underlying effect of the size factor, the argument could not be made that discharges or transfers occur solely due to policies which vary according to type of hospital or to differences in administration. Second, discharge rate represents an especially visible criterion and extrinsic measure of department performance, historically or comparatively (Jones and Sidebotham, 1962). This type of evaluative measure is also important to the inpatient psychiatric department because knowledge of treatment technology is incomplete (Perrow, 1965). In effect, then, assessment of effectiveness or efficiency has usually been made in an indirect way (Eaton, 1962; Perrow, 1961).

The other two measures of performance are related to cost of care. All expenditures directly attributable to the inpatient psychiatric service were totaled for the year 1969, including direct salary expenses, expenditures for equipment needed for inpatient care, and other operating expenditures, excluding overhead. For the year 1969, direct salary expenditures in all departments accounted for approximately 80 percent of the reported cost of inpatient care, which is not surprising in the light of the labor intensity of the psychiatric department. Total costs were then divided by (1) the total number of discharges from the inpatient department for the year, with the resulting quotient representing a cost per discharge index, and (2) the number of patient days, to arrive at a cost per patient day. The latter cost index is employed partly as a precautionary measure, since cost per discharge and discharge rate share a common figure, namely, number of discharges, and thus are not independent of one another. The number of patient days is also related to discharge rate, but in a more indirect way; that is, where more than one patient occupies a bed in a given day, through discharge and admissions on the same day, more than one patient day is counted. This suggests that the higher the discharge rate, the greater the probability of multiple census per bed. Consequently, the number of patient days as reported by the psychiatric departments is slightly inflated, and the cost per day slightly deflated, as a function of rate of discharge.

Professional and administrative staff. The Inventory of Psychiatric Services listed the total number of staff man-hours, full time and part time, worked for the inpatient psychiatric department during the week of January 11, 1970, the data-reporting period. Although the man-hours of weeks in 1969 might have varied somewhat from those of the data-reporting period, it was assumed that the variance was not sufficiently significant to affect results. Also, the fact that all departments reported manpower figures for the same week adds consistency to the inventory data.

Man-hours were listed by type of staff, denoting specifically

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such totals as the number of hours of psychiatrists, nurses, psychologists, and administrative personnel for the psychiatric department. Dividing total hours for each staff position by department size provided a measure of hours per unit of size or, similarly, a ratio of type of staff to size. Adjusting for size allows for comparisons among the 338 departments and for studying relative changes in professional and administrative staffing patterns after controlling for the absolute changes associated with variations in department size.

Input and output buffers. The existence of input buffers is determined by whether the psychiatric department had attempted to identify homogeneous elements of its task environment and restrict input accordingly. The Inventory of Psychiatric Services indicated whether a department had restrictions as to type of patient served. Examination for the existence of diagnostic restrictions reveals no differences in their occurrence as a function of hospital ownership (3 d.f., $\chi^2 = 3.42$, not shown). The types of restriction, of course, may still vary, but limitations in the Inventory data allow only the use of the existence of diagnostic restrictions. By such restrictions, the psychiatric department can become increasingly specialized, and thus more standardized in some of its responses. For efficiency, therefore, a psychiatric department in a heterogeneous environment might be expected to control admissions and selection procedures so as to encourage specialization and standardization. The "batching" phenomenon in mental hospitals (Goffman, 1957) is consistent with this contention of standardization, specialization of activities, and identification of homogeneous input elements.

The existence of an outpatient department or service suggests buffer activities for both input and output. Although the outpatient service has its own patients, the inpatient department occasionally receives patients for whom outpatient care is inappropriate. In this instance, the outpatient department is acting as an input buffer for the hospital. Some outpatient departments provide a follow-up service; this helps smooth output from the inpatient department by allowing for earlier release and providing effective extra-departmental care. Because the psychiatric inventory data did not specifically indicate which outpatient departments provided follow-up services, the existence of outpatient services was taken as a mixture of input and output buffers.

RESULTS

As stated in Hypothesis 1, it was expected that the performance of the psychiatric departments varied as a function of general hospital ownership. Table 1 shows that for the most part this hypothesis is supported. Comparison of discharge rates across type of hospital reveals significant differences ($F = 24.75$, $p < .001$). Most of the variance here appears to be accounted for by departments in Veterans Administration hospitals which have the lowest rate of discharge of all the departments examined. The high discharge rate of departments in city and county hospitals is the only finding contrary to expectations. Examination of both cost per discharge ($F = 8.12$, $p < .001$) and cost per patient-day ($F = 3.86$, $p < .01$) reveals results consistent with Hypothesis 1, with departments in private hospitals showing lower costs in both cases than the departments of government hospitals. As the data indicate, however, the divergence between the public and private departments is much smaller on cost per patient day than it is on cost per discharge.

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Table 1

Hospital Ownership and Performance and Size of Department							
Ownership		Department size * * * *					
	N	Average Discharge Rate *	Average Cost per Discharge **	Average Cost per Patient-Day ***	Small (8–24 beds)	Medium (25–40 beds)	Large (41 or more beds)
City/county	47	21.09	\$ 948.	\$34.22	29.8%	34.0%	36.2%
Veterans Administration	43	8.93*	1038.†	25.36‡	4.7	23.3	72.1
Private, profit-making	76	17.95	389.	21.96	34.2	39.5	26.3
Private, nonprofit	172	18.00§	529.	22.61#	37.2	37.8	25.0

* 3/334 d. f. (degrees of freedom), $F = 24.75$, $p < .001$
** 3/334 d. f., $F = 8.12$, $p < .001$
*** 3/334 d. f., $F = 3.86$, $p < .01$
**** 6 d. f. $X^2 = 38.84$, $p < .001$
* 1/88 d. f., $F = 44.13$, $p < .001$
† 1/88 d. f., $F = 1.07$, not significant at $p = .05$
‡ 1/88 d. f., $F = 4.40$, $p < .05$
§ 1/246 d. f., $F = .02$, not significant at $p = .05$
|| 1/246 d. f., $F = 2.16$, not significant at $p = .05$
1/246 d. f., $F = .81$, not significant at $p = .05$

Table 1 also shows that there are differences between the two types of government hospital. Consideration of discharge rate ($F = 44.13$, $p < .001$) and cost per patient day ($F = 4.40$, $p < .05$) indicates that departments in city and county hospitals perform much better on rates of discharge, but at a higher cost per patient day, than their counterparts in Veterans Administration hospitals. Contrary to these findings, there are no differences between departments in profit and nonprofit private hospitals on any of the three performance measures.

The differential constraints facing public and private inpatient psychiatric departments might be accounting for some of the observed differences in costs. The costs of departments in private hospitals are subject to scrutiny by both the hospital and the client paying for psychiatric services. Without the security of government support and the resulting absorption of financial uncertainty, the private department might be forced to maintain low costs in order to remain competitive and keep clients with the ability to pay for inpatient care. Furthermore, in the private organization some of the staff costs, for example, for physicians, are borne by the client through prepaid medical coverage, while in the Veterans Administration hospitals, the hospital usually pays staff costs in full. In the city and county hospitals, there is a mixture of cost coverage; the client may pay for some services directly. It is impossible to determine percentages from the present data, but the effects on costs of constraints associated with ownership must be recognized.

With regard to discharge rate, the data could indicate the relative inefficiency of psychiatric departments of the Veterans Administration hospitals compared to those in other types of hospitals, or there could be differences in the severity of illnesses treated by the Veterans Administration departments as compared to the other departments examined. Although it has been shown that the existence of diagnostic input restrictions does not vary by hospital ownership, potential differences in the types of restriction make it impossible to rule out the second explanation of differences in rates of discharge.

Consistent with Hypothesis 2, Table 1 indicates that size of the inpatient psychiatric department is related to hospital ownership, departments in government hospitals being larger than departments in private hospitals. Over 72 percent of Veterans Administration hospitals are classified as large, and less than 5 percent fall into the smallest size category. Departments in city and county hospitals are the next largest, while the departments of the private hospitals are found primarily in the small and medium-sized categories.

Although it has been shown that hospital ownership is differentially related to department performance and department size, the possibility that size affects performance negatively cannot be ignored. The low rate of discharge and the high cost per discharge of departments in Veterans Administration hospitals, for example, might be partly due to their large size and not only to hospital ownership, and the high level of performance of departments in private hospitals a function of size, not only private ownership. In order to ascertain the relationship between department size and performance, a correlation analysis was performed for each type of inpatient department (Table 2). Controlling for hospital ownership allows for the isolation and discussion of size effects.

Table 2

Correlation Analysis: Size and Performance of Department by Hospital Ownership				
Performance	Government		Private	
	City/County (N = 47)	V/A (N = 43)	Profit (N = 76)	Nonprofit (N = 172)
Discharge rate	-.34 *	-.40 **	-.24 *	-.29 **
Cost per discharge	.35 *	.49 **	.12	.18 *
Cost per patient day	.29 *	.31 *	.09	.15 *

* $p < .05$, one-tailed test
** $p < .01$, one-tailed test

Consistent with Hypothesis 4, it is seen that department size is negatively associated with department output, regardless of hospital ownership. Similarly, in all departments except those in private, profit-making hospitals, department size is positively related to cost per discharge and cost per patient day. These data suggest that increases in the size of the inpatient department are marked by various characteristics or changes, structural or otherwise, which decrease its ability to treat and discharge clients efficiently. In departments of private profit-making hospitals, increases in size do not result in increasing costs, implying again the salience of income and expenditures to those departments.

These findings do not negate those ascertained previously on the relationship between hospital ownership and department performance. Table 3 shows that hospital ownership has a significant impact on department discharge rate ($F = 8.68, p < .001$), cost per discharge ($F = 5.56, p < .01$), and cost per patient day ($F = 3.01, p < .05$), even after adjusting for the effects of department size. Thus, there are similarities among departments with respect to changes in department size, which consistently affect department performance. Equally clear is the fact that there are differences in performance as a function of hospital ownership.

The correlations in Table 4 showing the relationships between department size and department characteristics according to

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Table 3

Analysis of Covariance: Hospital Ownership and Department Performance, Controlling for Effects of Department Size				
Performance •	City/county	Veterans	Profit	Nonprofit
	(N = 47)	(N = 43)	(N = 76)	(N = 172)
Discharge rate	21.22	11.75	17.63	17.60
Standard error	9.23	4.02	5.55	6.97
F-Ratio (d.f. 3/333) = 8.68, $p < .001$				
Cost of discharge	\$ 939	917	407	544
Standard error	1052	802	292	493
F-Ratio (d.f. 3/333) = 5.56, $p < .01$				
Cost per patient day	\$33.60	23.91	22.10	23.00
Standard error	30.45	16.79	16.01	19.76
F-Ratio (d.f. 3/333) = 3.01, $p < .05$				
• Adjusted				

Table 4

Correlation Analysis: Size and Characteristics of Departments by Hospital Ownership				
Department variables	Government	Private		
	City/County (N = 47)	Veterans Administration (N = 43)	Profit (N = 76)	Nonprofit (N = 172)
Administrative staff*	-.04	.12	-.06	-.07
Professional staff*				
Psychiatrists and other physicians	.31 •	.32 •	-.09	-.02
Psychologists	.29 •	.32 •	-.13	.11
Other professionals (excluding nurses)	.04	.11	.24 •	.26 ••
Nurses, paraprofessionals, attendants involved in direct care	-.10	-.34 •	.23 •	.16 •
Diagnostic restrictions	.29 •	.34 •	-.03	.17 •
Outpatient services	.42 ••	.30 •	-.12	.19 •
• $p < .05$ (one-tailed test)				
•• $p < .01$ (one-tailed test)				
*Man-hours per bed				

hospital ownership indicate similarities and differences among departments, and provide an insight into the adaptive behavior and rationality of the psychiatric department and their relation to hospital ownership. For example, it is seen that increases in department size are not marked by changes in the size of administrative staff but, consistent with Hypothesis 3, there are changes in the relative size of the professional or semi-professional staff. The professionalization of staff, however, differs in government and private hospitals. In the government hospitals the greatest increases in professional staff are found among psychiatrists and other physicians and psychologists, with decreases among lesser-paid professional

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Table 5

Means and Standard Deviations for Department Staff Variables by Department Size and Hospital Ownership

Staff variables	City / county				Veterans Administration			
	Small	Medium	Large	Average	Small	Medium	Large	Average
Psychiatrists & other physicians	2.41 • (4.30)†	4.16 (4.19)	4.80 (4.06)	3.85 (4.19)	1.60 (.57)	3.65 (2.59)	4.27 (3.37)	3.98 (2.71)
Psychologists	1.74 (2.28)	2.60 (2.06)	3.04 (2.95)	2.46 (2.45)	2.35 (.49)	3.53 (1.99)	4.42 (2.55)	4.04 (2.52)
Other professionals (excluding nurses)	2.31 (1.82)	2.48 (2.36)	2.87 (3.47)	2.51 (2.69)	1.75 (.35)	.81 (.63)	1.93 (1.73)	1.67 (1.56)
Nurses, paraprofessional staff involved in direct patient care	27.11 (10.30)	24.69 (5.04)	26.91 (11.05)	25.94 (9.29)	19.70 (2.99)	19.54 (6.78)	17.44 (5.93)	17.70 (6.04)
Administrative staff	.24 (.71)	.24 (.28)	.37 (.65)	.27 (.57)	.75 (1.07)	1.49 (2.34)	1.68 (3.06)	1.58 (2.48)

• Numbers without parentheses represent man-hours per bed.

†Numbers in parentheses represent standard deviations.

and paraprofessional employees; in private hospitals increases in department size are marked by a proliferation of the lower priced professional and paraprofessional staff. The government-controlled departments clearly experience the greater professionalization of staff and the larger increase in labor costs.

The means and standard deviations in Table 5 for staff man-hours per bed by hospital ownership and department size show more clearly the greater emphasis on physicians and psychologists in the departments of government hospitals compared to the departments of private hospitals. The man-hours worked in the private hospitals do not include hours of those physicians and, to a lesser extent, psychologists, who are not paid by the hospital and who only use hospital facilities intermittently; that is, actual physician and psychologist man-hours are somewhat understated. Discussions with personnel at the National Institute of Mental Health, however, indicate that the discrepancy between reported and actual man-hours is probably not very large.

The relative paucity of nursing hours per bed in the departments of Veterans Administration hospitals is especially interesting compared to the same figure in departments of city and county hospitals. The percentage of total staff hours accounted for by man-hours of nurses and paraprofessional employees is much higher in the profit-making (87.9 percent) and nonprofit (83.3 percent) private hospitals than in the public city and county (74.1 percent) and Veterans Administration (61.0 percent) hospitals. The emphasis on administrative personnel in departments of the Veterans Administration hospitals relative to their employment in all of the other departments is another point of difference shown in Table 5.

Differences in staffing patterns are emphasized if one considers changes in staff man-hours as a function of department size. Table 5 shows the relative increases in highly paid professional staff in the government-controlled hospitals and the greater employment of nurses and paraprofessional staff in the private hospitals associated with increases in size. These data indicate that the variation of staffing patterns with department size is a function of hospital ownership.

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Table 5 (continued)

Staff variables	Private, profit				Private, nonprofit			
	Small	Medium	Large	Average	Small	Medium	Large	Average
Psychiatrists & other physicians	.43 (.95)	.63 (1.45)	.72 (1.57)	.59 (1.31)	1.62 (2.45)	1.34 (2.37)	1.55 (2.88)	1.48 (2.88)
Psychologists	.64 (.97)	.73 (1.33)	.66 (.98)	.69 (1.12)	.97 (1.45)	1.20 (1.71)	1.44 (2.36)	1.20 (2.11)
Other professionals (excluding nurses)	1.94 (2.27)	2.12 (1.72)	2.78 (1.88)	2.24 (1.96)	1.43 (1.51)	1.94 (1.46)	2.71 (2.05)	2.03 (1.91)
Nurses, paraprofessional staff involved in direct patient care	26.52 (10.00)	27.98 (7.23)	29.82 (7.21)	28.10 (9.78)	26.32 (7.04)	27.45 (8.08)	28.57 (9.41)	27.01 (8.21)
Administrative staff	.20 (.60)	.51 (1.02)	.28 (1.03)	.33 (1.01)	.65 (2.93)	.78 (2.95)	.66 (1.50)	.69 (2.63)

Consistent with Hypothesis 5, the existence of input and output buffers is directly related to department size in all departments except those in private profit-making hospitals. Increases in size are correlated with the increased specialization of staff and the development of structural subunits to buffer the technological core of the inpatient departments in both public and private nonprofit hospitals (Table 4). On the input side, there is a positive relationship between the existence of diagnostic restrictions and department size, thus indicating an attempt to control admissions and encourage task standardization and specialization of activity. Similarly, the positive relationship between size and the development of outpatient services suggests an effort to surround the treatment technology with input and output buffers.

Multivariate Analysis

Analysis thus far has shown that hospital ownership and department size are differentially related to department discharge rate and costs. Tables 4 and 5 have indicated, however, that there are structural correlates of size and differences in staffing patterns by hospital ownership given changes in department size. One question that still remains is the relative importance of the resource inputs and structural factors for department performance. That is, it is necessary to consider whether department size alone affects rate of discharge and costs, or whether some of the apparent correlates of size have effects on performance independent of department size but dependent on hospital ownership.

To determine the relative importance of size and its correlates and to derive predictive equations for department performance, partial correlations and stepwise multiple regressions were performed. In addition to department size, input resources, including the dollar costs of patient care, were used as independent variables in an analysis of discharge rate. Department size, staffing patterns, and the existence of input and output buffers were employed as independent measures in an effort to explain cost per discharge and cost per patient day for the psychiatric departments. Independent effects can thus be shown, making it easier to recognize the importance of variables for department performance as well as the differences or similarities across hospitals.

Table 6 presents the results of the multivariate tests with discharge rate as the dependent variable. In all cases,

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Table 6

Partial Correlation and Regression Analysis: Dependent Variable = Discharge Rate

Independent variables	City/ county		Veterans Administration		Private, profit		Private, nonprofit	
	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²
Department size (S)	-.323 *	.104	-.542 ***	.294	-.252 *	.064	-.355 ***	.126
Psychiatrists and other physicians (M)	-.326 *	.106	-.363 **	.132	-.182	.033	-.182 *	.033
Psychologists (P)	-.165	.027	-.196	.038	.125	.016	.072	.005
Nurses, paraprofessionals (N)	.045	.002	.190	.036	.184	.034	.100	.010
Administrative staff	.124	.015	.079	.006	.054	.003	.068	.005
Diagnostic restrictions (A)	.059	.003	-.182	.033	.141	.020	.063	.004
Outpatients services (Q)	.034	.001	.045	.002	-.125	.016	.058	.003
Total direct expenses (E)	-.300 *	.090	-.098	.010	.016	.000	.059	.003
Y = discharge rate	Y = 29.72 - .046 - .650 - .0003‡		Y = 13.59 - .031 - .454§		Y = 19.83 - .053		Y = 19.97 - .017S - .831M#	Y = d

• p < .05
•• p < .01
••• p < .001

‡ 3.43 d.f., F = 2.852 •
§ 2.40 d.f., F = 12.582 •••
|| 1.74 d.f., F = 4.356 •
2.169 d.f., F = 14.564 •••

department size is negatively related to department output in terms of discharge rate. The variance explained by size ranges from 6.4 percent for the private profit and 12.6 percent for the nonprofit hospitals to 10.4 percent for the government city-county hospitals and 29.4 percent for Veterans Administration hospitals. These data thus reinforce the preceding data, showing the negative relation of department size to rate of patient discharge.

The input factor, number of hours of psychiatrists and physicians, shows effects that are independent of department size but at least partly related to organizational ownership. Analysis of the partials for the departments in city and county hospitals, for example, shows this input variable as having the most significant effect, accounting for 10.6 percent of the variance in output. Although the effect of the physician input variable is secondary to that of department size in the Veterans Administration hospitals, it still is very significant, explaining over 13 percent of the variance in discharge rate. In contrast, the effects of the physician variable are insignificant for departments in private, profit-making hospitals. In the private, nonprofit hospitals, this input factor is statistically significant, although the proportion of total variance explained is really too small to be very meaningful ($R^2 = .033$). The relationship between discharge rate and the input factor, number of psychiatrists and physicians, therefore, is more striking for the departments in public hospitals than for the departments in private hospitals.

The only other independent effect shown in Table 6 is the variable, total direct expenses, for the departments in

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city-county hospitals. Thus the regression equation predicting discharge rate for those departments uses three independent variables. For the departments of Veterans Administration and nonprofit hospitals, the regression equation includes department size and the physician input variable. In profit-making hospitals, the only variable sufficiently significant to enter the regression equation is department size; rate of discharge can be predicted simply given knowledge of that one departmental characteristic. In the light of the relationships in Table 4 and the partial correlations in Table 6, holding the effects of size constant, it can be concluded that most independent variables are size correlates without separate effects on rates of discharge.

In all cases, tests for the existence of regression prove positive, with F-ratios of 2.852 ($p < .05$) for the city and county hospitals, 12.582 ($p < .001$) for Veterans Administration hospitals, 4.356 ($p < .05$) for the profit-making hospitals, and 14.564 ($p < .001$) for the nonprofit hospitals.

The regression analysis with cost per discharge as dependent variable (Table 7) shows that for departments in the government hospitals, department size is the most important explanatory variable, the variance explained by department size being 13.9 percent in the city and county hospitals, and 27.2 percent in the Veterans Administration hospitals. Man-hours per bed of psychiatrists and other physicians ($R^2 = .110$), as well as nurses and paraprofessional employees ($R^2 = .091$), are significantly related to cost per discharge in departments of city and county hospitals. The existence of

Table 7

Partial Correlation and Regression Analysis: Dependent Variable = Cost per Discharge								
Independent variables	City/county		Veterans Administration		Private, profit		Private, nonprofit	
	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²
Department size (S)	.373 ••	.139	.522 •••	.272	.038	.001	.122	.015
Psychiatrists and other physicians (M)	.331 •	.110	.215	.046	.116	.013	.444 •••	.197
Psychologists (P)	.266	.071	.124	.015	.269 •	.072	.218 ••	.048
Nurses, paraprofessionals (N)	.301 •	.091	-.198	.039	.365 ••	.133	-.149 •	.022
Other professional staff	.074	.005	.153	.023	.090	.008	-.085	.007
Administrative staff (A)	.047	.002	.283	.81	.364 ••	.132	.088	.008
Diagnostic restrictions	.137	.019	.072	.005	-.097	.009	.055	.003
Outpatient services (Q)	.114	.013	-.304 •	.092	.202	.041	.062	.004
Cost per discharge	CD = 68.66 + 22.225S + 121.89M + .12N‡		CD = 834.06 + 5.74S - 493.37Q§		CD = 177.46 + 5.32N + 1.21A + 58.28P		CD = 187.81 + 124.28 + 73.64P + 2.43N#	
• <i>p</i> < .05	‡ 3/72 d.f., <i>F</i> = 3.096 •							
•• <i>p</i> < .01	§ 2/40 d.f., <i>F</i> = 8.478 ••							
••• <i>p</i> < .001	3/72 d.f., <i>F</i> = 6.672 ••							
	# 3/168 d.f., <i>F</i> = 28.969 •••							

outpatient services ($R^2 = .092$) is the only significant variable besides department size in the analysis of cost per discharge in the Veterans Administration hospitals.

Analysis of cost per discharge for the private hospitals shows that department size is not significant, whereas it was the most important explanatory measure in both types of government-controlled hospitals. Man-hours of nurses and paraprofessionals, administrative staff, psychiatrists, and psychologists are significantly related to cost per discharge in the private hospitals. In the analysis of cost per discharge, therefore, the nurse-paraprofessional variable appears significant in all cases except for the Veterans Administration hospitals, the physician variable in both the city and county and the private nonprofit hospitals, the department size variable is specific to government hospitals, while the psychologist variable appears significant only in the private hospitals. Tests for the existence of regression of cost per discharge all prove significant.

Table 8

Partial Correlation and Regression Analysis: Dependent Variable = Cost per Patient Day								
Independent variables	City/county		Veterans Administration		Private, profit		Private, nonprofit	
	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²	Partial R	Partial R ²
Department size (S)	.292 *	.085	.306 *	.094	.107	.011	.122	.015
Psychiatrists and other physicians (M)	.376 **	.141	.267	.071	.170	.029	.377 ***	.142
Psychologists (P)	.205	.042	.233	.054	.193	.037	.120	.014
Nurses, paraprofessionals (N)	.314 *	.099	-.126	.016	.504 ***	.254	.206 **	.042
Other professional staff	.063	.004	.164	.027	.059	.003	.103	.011
Administrative staff (A)	.045	.002	.063	.004	.249 *	.062	.146 *	.022
Diagnostic restrictions	-.097	.009	-.267	.071	.063	.004	.075	.006
Outpatient services (Q)	.114	.013	-.341 *	.116	.114	.013	-.095	.009
Cost per patient day	CP = .342 + .129S + 2.12M + .67N‡		CP = 30.95 - 10.11Q + .004S§		CP = -6.45 + .980N + .045A		CP = 7.90 + 2.45M + .468N + .023A #	
• $p < .05$	‡ 3/43 d.f., $F = 4.399$ **							
•• $p < .01$	§ 2/40 d.f., $F = 2.832$ •							
••• $p < .001$	2/73 d.f., $F = 17.458$ •••							
	# 3/168 d.f., $F = 18.674$ •••							

Examination of the partial correlation and regression analysis for cost per patient day (Table 8) shows results very similar to those found when considering cost per discharge. While there may be some changes in the relative importance of variables as suggested by the proportion of variance explained, for the most part the same variables appear and the same relationships hold. Department size, for example, appears significant in the regression analysis for the city and county ($R^2 = .085$) and Veterans Administration ($R^2 = .094$) hospitals, but is not important for cost per patient day in either type of private hospital. The effects of size, then, are again specific to

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the government hospitals, having no significant impact on costs in the private hospitals.

Other explanatory measures for cost per patient day whose appearance in Table 8 is consistent with that when considering cost per discharge include: the physician input factor in the city and county ($R^2 = .141$) and nonprofit ($R^2 = .142$) hospitals; the nurse and paraprofessional staff variable, significant in all except the Veterans Administration hospitals ($R^2 = .099$, .254, and .042 for the city and county, profit, and nonprofit hospitals, respectively); the man-hours of administrative staff ($R^2 = .062$) in private, profit hospitals; and the existence of out-patient services ($R^2 = .116$) in the Veterans Administration hospitals. One difference is the lack of significance of the psychologist variable for cost per patient-day in profit-making and nonprofit hospitals. That there should be a greater number of similarities than differences in the multivariate tests for cost per discharge and cost per patient day is suggested by the correlations between those two measures. In the city and county and Veterans Administration hospitals, the correlations between the cost variables are .761 and .538, respectively, while for the profit and nonprofit hospitals, they are .713 and .628, respectively.

Finally, as was the case with cost per discharge, tests for the existence of regression are positive in the departments of city and county ($F = 4.399$, $p < .01$), Veterans Administration ($F = 2.832$, $p < .05$), profit ($F = 17.458$, $p < .001$), and nonprofit ($F = 18.674$, $p < .001$) hospitals.

CONCLUSIONS AND DISCUSSION

In this study of 338 inpatient psychiatric departments in public and private general hospitals, five hypotheses were tested and supported with the following results:

- (1) The performance of the inpatient psychiatric departments varied with hospital ownership. With one exception, departments in private hospitals performed better than those in government-owned hospitals when considering rate of patient discharge, cost per discharge, and cost per patient day. Departments of city and county hospitals showed a higher rate of patient discharge and higher costs per patient day than departments in Veterans Administration hospitals, but no differences were found between the two types of private hospital departments on any of the performance measures.
- (2) The smallest inpatient psychiatric departments were found predominantly in private general hospitals, with the largest departments primarily in government-owned hospitals.
- (3) Size of inpatient psychiatric departments was not related to changes in the relative size of administrative staff, but was associated with increases in professional or semi-professional staff. Staff increases were noted among physicians and psychologists in government hospitals, and in the nurses and paraprofessional staff in private hospitals.
- (4) Size was negatively related to rate of discharge in all departments and positively related to cost per discharge and cost per patient day in government-owned hospitals. Size was not related to either of the cost measures in the private hospitals.
- (5) Increases in size were correlated with attempts to buffer the technology of inpatient care. That is, the existence of diagnostic input restrictions and outpatient units or services was positively related to department size.

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The findings have implications for system development and performance. Increasing size seems to affect organizational performance adversely. In all departments, increased size showed a significant negative relation to rates of patient discharge, suggesting that size increases might accentuate problems such as those related to coordination and communication. This is partly supported by earlier findings in large mental hospitals of poor staff communication and a lack of integration of professional activities which, in turn, affected client-staff interaction adversely and inhibited the effective treatment of clients (Greenblat, et al., 1957; Goffman, 1961; Stanton and Schwartz, 1954; Cumming and Cumming, 1956 and 1962). Growth of the psychiatric department also could increase the probability of receiving patients who are chronically ill; thus, decreases in rates of discharge with increasing size could reflect changes in treatment emphasis and not simply the problems that tax the structure of the organization. Finally, in the government hospitals, which are large, many of the variables in the partial correlation analysis showed a negative relationship with discharge rate, indicating that size is an important variable for organizational performance. If the effects of size are ubiquitous and far-reaching, as Meyer (1972) indicated, perhaps even the proliferation of professional staff could not counteract the negative impact of size and improve performance. In the private hospitals, which had smaller departments, there were few negative relationships with discharge rate after controlling for size, further emphasizing the effects of increasing size.

Organizational ownership was found to be related to department performance. In the analysis of cost per discharge and cost per patient-day, the variables entering the regression equations were determined largely by organizational ownership. Size, for example, was a significant factor for both cost measures only in the departments of government hospitals.

Strategies for patient treatment seemed to differ as a function of organizational ownership, as evidenced by the increases in highly paid professional staff in government hospitals and in the number of semi-professional staff in private hospitals, with increases in department size. This could reflect either differences in the use of staff or staff specialization or differences in concern with costs. The high staff specialization in government departments, for example, could reflect an adherence to formal policy and government personnel charts, and might not reflect actual duties performed. On the other hand, levels of cost may be more salient to private hospitals. It has been observed that levels of financial uncertainty, expectations of sponsors, competition, and various budgetary controls vary with organizational ownership (Thompson, 1967; Elling and Halebsky, 1961; Blau and Scott, 1962). However, some of the costs in the private hospitals, for example, for psychiatrists or other physicians, are borne by the patient; this could bias the cost data in favor of the private hospital. The data on staff man-hours by department size and ownership of hospital, coupled with the staff changes with increases in department size, nonetheless suggest that there are basic differences between government and private hospitals. Furthermore, in the analysis of cost per patient-day, department size was the most significant factor for the government hospitals but was not significant for private hospitals, suggesting again important differences as a function of ownership other than those associated with the use of highly-paid professional staff.

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The establishment of buffer mechanisms for input and output with increasing size suggests attempts to regulate and standardize the flow of work through the psychiatric department. One issue not explored in this paper is the extent to which boundary units in turn develop buffers as they grow in size. The existence of neighborhood clinics associated with admissions units and half-way houses attached to outpatient units suggests that they experience similar centrifugal tendencies and buffering activities.

The inverse relationship between man-hours of physicians and the discharge rate of departments in government hospitals (Table 6) might suggest that professional expertise is being used to reduce ambiguity and enhance legitimacy in a technology in which the criteria for performance are not well known. That is, there might be a tendency to employ highly-paid professionals even when it is inappropriate to do so. If professionals are asked to perform nonprofessional tasks, they may become dissatisfied, leading to such unfavorable consequences as reductions in commitment (Hrebiniak and Alutto, 1972) and increased turnover. However, only when there is a basis for comparing costs and environmental pressure to reduce costs will such systems be able to consider reassignment of personnel. This suggests that comparative organizational analysis may generate not only more efficient systems but greater satisfaction and fulfillment for personnel.

Finally, one should remember that measures of effectiveness such as rates of recidivism and readmission were not used in this study; thus the issue of the relationship between subsystem size, organizational setting, and actual product value remains a question in need of additional research.

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